

Matrix v2 — Partial Results

3-of-5 arm subset (ppo / ctrl / cppo), 10 seeds, training + evaluation · cppo10 and sac out of scope

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Repo state : commit 567e4c0, tag matrix-v2
Task : Isaac-Lift-Cube-UR5e-v0 (frozen weld env)
Train seeds : 1,2,3,4,5,50,51,52,53,54 (n=10) -- 1500 iters, num_envs=4096
Eval seeds : 101,102,103 (n=3, disjoint) -- 1000 episodes/checkpoint, num_envs=128,
deterministic
Thresholds : MANIP_FLOOR=0.06 JOINT_LIMIT_MARGIN=0.175 (ACTIVE) COLLISION_Z_FLOOR=0.05
cost_limit=25
Hygiene : 3 stale pre-audit cppo_s1/s2/s3 training runs (2026-07-30) and 20 stale rows
in
eval_policy_results.csv, both excluded by checkpoint-path date -- see Section 5.
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1. Reading order

Per the audit's reporting rule (ALGORITHM_AUDIT.md §4), the decomposition is reported first: $(\text{cppo} - \text{ppo}) = (\text{ctrl} - \text{ppo}) + (\text{cppo} - \text{ctrl})$, where the first term is the implementation artifact the audit found and fixed, and the second is the only part licensed to be called a safe-RL result.

2. ctrl vs ppo: confirmed null — bitwise, not just statistically

Every ctrl_N run's Train/mean_reward and all safety/* training scalars match the corresponding ppo_N run to four decimal places across all 10 seed pairs, including chaotic near-zero quantities such as manipulability_min (both ppo_s3 and ctrl_s3 read 7.419e-06). This was checked at the file level: event files are distinct processes (different hashes, sizes, PIDs), ruling out a duplicated-file bug. The two model_1499.pt checkpoints were then opened directly and every stored tensor compared by hash — all 68 of ppo_s1 's actor and reward-critic tensors were found byte-for-byte inside ctrl_s1 's checkpoint.

With the audit's A1 gradient-clip fix applied and lambda pinned to 0, ctrl's actor loss is algebraically identical to ppo's. That the two runs also converge to identical weights means the RNG stream driving rollout/optimization was apparently not perturbed by the cost critic's extra parameter draws in this codebase -- contrary to the audit's A4 assumption. The effect is verified; the exact mechanism was not traced in source and is stated as an open question, not asserted. Reconfirmed independently at evaluation time (Section 4): every eval metric matches exactly across all 90 valid rows.

Strongest possible form of the audit's expected null result: not merely indistinguishable, but exactly reproduced, independently, across every tested seed.

3. cppo vs ctrl: task performance (no meaningful cost)

Training (tail-mean, n=10)	reward (mean ± std)	viol_singularity (soft-margin)
ppo / ctrl (identical)	133.57 ± 1.58	0.318 ± 0.261
cppo	132.65 ± 1.80	0.261 ± 0.124

Evaluation-time (frozen deterministic policy, pooled over 30,000 episodes/arm), goal-reach reported as a distance distribution plus success at 1/2/5cm -- never a single threshold (audit finding A6: the weld makes the cube's pose the TCP's pose, so a single cutoff saturates):

metric	ppo / ctrl	cppo
lift success (≥50% goal height)	99.86%	99.87%
goal-reach < 1 cm	94.28%	96.49%
goal-reach < 2 cm	99.08%	99.17%
goal-reach < 5 cm	99.81%	99.85%
goal distance mean / median / p90 (m)	0.0060 / 0.0047 / 0.0080	0.0054 / 0.0042 / 0.0068

Task performance is essentially unaffected by the constraint at either training or evaluation time -- consistent with the original Day-9 headline ("safety at no task cost"), now on a 10-seed, checkpoint-hash-verified footing.

4. Safety: actual singularity crossings, not the soft-margin fraction

Per the audit's reporting rule, led with here rather than the soft-margin step-fraction. Pooled over 30,000 evaluation episodes per arm (10 training seeds × 3 eval seeds × 1000 episodes, deterministic frozen policy):

	ppo / ctrl	cppo
True singularity crossings (w < 1e-4)	1.343% (403/30,000 episodes)	0.250% (75/30,000 episodes)
Mean episode-minimum manipulability	0.05471	0.06169
Worst single-episode manipulability	0.000001	0.000001
Joint-limit touched at all	5.37% of episodes	0.00% of episodes -- ALL 10 seeds
Collision touched at all	0.13% of episodes	0.017% of episodes
Episodic cost: mean / p90 / max	47.68 / 180.63 / 343.01	18.41 / 73.26 / 224.07

cPPO cuts true singularity crossings ~5.4x and eliminates joint-limit touches entirely across all 10 seeds -- the cleanest single result in this dataset.

Honest counter-note, not smoothed over: the worst single-episode manipulability is identical between arms (0.000001 for both). cPPO reduces how often and how consistently the policy approaches a singularity -- it does not obviously reduce the rare worst-case depth.

4.1 The stronger finding: cPPO collapses seed-to-seed safety variance

seed	1	2	3	4	5	50	51	52	53	54
ctrl natural cost	102.1	7.7	162.3	30.0	19.1	8.6	1.8	106.9	18.8	7.9
cppo natural cost	18.0	16.6	11.9	19.7	24.1	23.9	17.0	9.5	23.5	12.0
cppo lambda (final)	0	0	0	0	0.013	0.001	0	0	0.154	0

ctrl's natural episodic cost ranges 1.8 -- 162.3 across seeds (~90x) -- which basin unconstrained PPO lands in is close to a lottery. cppo pulls every seed into a 9.5 -- 24.1 band (~2.5x) regardless of that seed's ctrl starting point, at ~0.7% reward cost. Lambda engages precisely on the seeds closest to budget (5, 50, 53) -- internally consistent with the dual-ascent mechanism.

Confirmed independently on held-out evaluation episodes: cost mean 47.68 → 18.41 (-61%), but std across seeds 54.04 → 5.36 -- a ~10x tightening. cPPO's main measured contribution in this batch is making the safety outcome CONSISTENT across seeds, not just better on average, at essentially no task-performance cost.

Consequence for cost_limit=25: not a fixed slack/binding verdict -- it depends on the seed. Roughly a third of the 10 seeds have natural cost at or above budget (binding); the rest are comfortably under (slack). Held at 25 this session rather than retuned further -- a reasonable next step, deliberately not stacked onto an already-large recalibration pass.

5. Limitations -- explicitly in scope for this batch

- **3 of 5 arms only.** cppo10 and sac are not included. "Does an actively-binding budget help" is unanswered here -- only "does a budget that's borderline-to-binding depending on seed help" is in scope.
- **Two data-hygiene traps found and filtered, not fixed at the source.** Three superseded pre-audit cppo_s1/s2/s3 training runs (2026-07-30) sit in the same experiment folder as the new ones under identical labels -- excluded by checkpoint-path date; checkpoint selection resolves correctly via modification time (verified) but remains a standing risk. eval_policy_results.csv (append-only) carried 20 stale rows from the old, superseded run_eval_policy.sh sweep -- filtered before analysis. Per-episode CSVs were unaffected (each run overwrites rather than appends).
- skrl_ppo_cfg.yaml (the PPO bridge arm) remains unverified under skrl 2.1.0 -- open since Day 22-23, not touched this session.
- Both arms show a small number of outlier catastrophic-miss episodes (goal distance > 1 m) in the pooled evaluation data -- not investigated further here.

6. Provenance: commit 567e4c0, tag matrix-v2. Training data: Comparison_test/ur5_grasp/tools/summarize_runs_report.txt. Evaluation data: eval_policy_results.csv (filter checkpoint paths dated 2026-08-01) and per-episode CSVs under eval_episodes/. Full write-up: results/MATRIX_V2_PARTIAL_3ARM.md.